Stack & Queue MCQs (1-50)

Q1. Which data structure is used for recursion?
A) Queue
B) Stack
C) Linked List
D) Tree
Answer: B

Q2. Top after POP, POP, PUSH(50) on stack 10,20,30,40(top)?
A) 20
B) 30
C) 40
D) 50
Answer: D

Q3. Time complexity of PUSH and POP in linked-list stack?
A) O(n),O(n)
B) O(1),O(1)
C) O(logn),O(logn)
D) O(n),O(1)
Answer: B

Q4. Stack overflow occurs when:
A) Pop empty
B) Push full
C) Peek
D) Traverse
Answer: B

Q5. Balanced parentheses use:
A) Queue
B) Stack
C) Heap
D) Graph
Answer: B

Q6. Which notation needs no parentheses?
A) Infix
B) Prefix
C) Postfix
D) Both Prefix & Postfix
Answer: D

Q7. Queue follows:
A) LIFO
B) FIFO
C) FILO
D) Random
Answer: B

Q8. Queue insertion at:
A) Front

B) Rear
C) Middle
D) Anywhere
Answer: B

Q9. Queue deletion at:
A) Rear
B) Front
C) Middle
D) Anywhere
Answer: B

Q10. Major advantage of circular queue?
A) Speed
B) Efficient memory
C) Sorting
D) Recursion
Answer: B

Q11. BFS uses:
A) Stack
B) Queue
C) Heap
D) Tree
Answer: B

Q12. DFS uses:
A) Queue
B) Stack
C) Array
D) Hash
Answer: B

Q13. Queue underflow occurs on:
A) Enqueue full
B) Dequeue empty
C) Peek
D) Traverse
Answer: B

Q14. Stack implementation:
A) Array
B) Linked List
C) Both
D) Tree
Answer: C

Q15. TOP=3. After PUSH, TOP=?
A) 2
B) 3
C) 4
D) 5

Answer: C

Q16. Not a stack application?
A) Function calls
B) Parentheses
C) BFS
D) Undo
Answer: C

Q17. TOP=-1. After 3 PUSH, TOP=?
A) 1
B) 2
C) 3
D) 4
Answer: B

Q18. Traversal using stack?
A) Inorder
B) Preorder
C) Postorder
D) All
Answer: D

Q19. Find bottom of stack complexity?
A) O(1)
B) O(logn)
C) O(n)
D) $O(n^2)$
Answer: C

Q20. Browser Back uses:
A) Queue
B) Stack
C) Heap
D) Tree
Answer: B

Q21. Linear queue wastes memory due to:
A) Overflow
B) Underflow
C) Linear implementation
D) Linked list
Answer: C

Q22. Circular queue full when:
A) F=R
B) (R+1)%N=F
C) R=N
D) F=N
Answer: B

Q23. Impossible in standard queue?
A) Insert rear
B) Delete front
C) Delete rear
D) Peek
Answer: C

Q24. Linked queue enqueue complexity?
A) O(1)
B) O(logn)
C) O(n)
D) O(nlogn)
Answer: A

Q25. Printer scheduling uses:
A) Stack
B) Queue
C) Tree
D) Graph
Answer: B

Q26. Deque insertion:
A) Front
B) Rear
C) Both ends
D) None
Answer: C

Q27. No precedence required in:
A) Infix
B) Prefix
C) Postfix
D) Both B & C
Answer: D

Q28. Queues to implement stack:
A) 1
B) 2
C) 3
D) 4
Answer: B

Q29. Stacks to implement queue:
A) 1
B) 2
C) 3
D) 4
Answer: B

Q30. Peek returns:
A) Bottom
B) Top
C) Middle

D) Largest
Answer: B

Q31. Valid postfix:
A) AB+C*
B) A+B*C
C) (A+B)*C
D) A*(B+C)
Answer: A

Q32. Stacks for postfix evaluation:
A) 0
B) 1
C) 2
D) 3
Answer: B

Q33. Queue policy:
A) FIFO
B) LIFO
C) FILO
D) Random
Answer: A

Q34. Stack policy:
A) FIFO
B) LIFO
C) RR
D) Priority
Answer: B

Q35. Not queue operation:
A) ENQUEUE
B) DEQUEUE
C) PUSH
D) PEEK
Answer: C

Q36. Linked queue insertion at:
A) Head
B) Tail
C) Middle
D) Anywhere
Answer: B

Q37. Space complexity of stack:
A) O(1)
B) O(logn)
C) O(n)
D) $O(n^2)$
Answer: C

Q38. DFS mainly uses:
A) Queue
B) Stack
C) Heap
D) Graph
Answer: B

Q39. Queue serves:
A) Latest
B) First inserted
C) Random
D) Highest priority
Answer: B

Q40. Queue application:
A) CPU scheduling
B) Undo
C) Recursion
D) Expression eval
Answer: A

Q41. Stack overflow when:
A) TOP=-1
B) TOP=SIZE-1
C) F=R
D) R=0
Answer: B

Q42. Stack underflow when:
A) TOP=SIZE-1
B) TOP=-1
C) R=SIZE
D) F=R
Answer: B

Q43. Reverse string using:
A) Queue
B) Stack
C) Heap
D) Graph
Answer: B

Q44. Priority queue removes by:
A) Insertion
B) Priority
C) Alphabet
D) Random
Answer: B

Q45. Level order traversal uses:
A) Stack
B) Queue
C) Linked List

D) Heap
Answer: B

Q46. Queue using two stacks worst-case:
A) O(1)
B) O(logn)
C) O(n)
D) $O(n^2)$
Answer: C

Q47. Non-recursive DFS uses:
A) Queue
B) Stack
C) Heap
D) Array
Answer: B

Q48. TOP increases on:
A) POP
B) PUSH
C) PEEK
D) DISPLAY
Answer: B

Q49. Circular queue size8 F3 R6 enqueue new R?
A) 6
B) 7
C) 0
D) 3
Answer: B

Q50. Correct statement:
A) Stack FIFO Queue LIFO
B) Both FIFO
C) Stack LIFO Queue FIFO
D) Both LIFO
Answer: C